CYBERSECURITY INTERNSHIP

WEEK 5 ASSIGNMENT REPORT

VAPT TOOLS: PRACTICAL LEARNING

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Topics covered: Network Scanning — Nmap • Web Vulnerability Assessment — Nikto • HTTP Traffic Analysis — Burp Suite

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1. INTRODUCTION & VAPT OVERVIEW

About This Assignment

This practical assignment demonstrates a basic Vulnerability Assessment and Penetration Testing (VAPT) workflow through hands-on testing of an authorized training website. The work focuses on network discovery, web-server assessment, HTTP traffic interception, and documentation of observed results.

What is VAPT?

- Identify exposed network services and understand how scanning tools report them.
- Assess web-server security using a dedicated vulnerability-scanning tool.
- Intercept and inspect HTTP/HTTPS traffic to understand client-server communication.
- Document findings based on actual tool output and scan evidence.
- Practice security testing only against authorized targets.

Target and Scope

The supplied presentation identifies testphp.vulnweb.com as the authorized training target. The practical work covers Nmap, Nikto, and Burp Suite.

2. OBJECTIVES AND VAPT WORKFLOW

Objective

To understand the basic Vulnerability Assessment and Penetration Testing process through hands-on testing of an authorized training website.

VAPT Workflow

Target Identification → Network Scanning → Web Assessment → Traffic Analysis → Finding Documentation

Practical Objectives

- Perform network scanning using Nmap.
- Identify open ports and available services.
- Perform web-server assessment using Nikto.
- Intercept HTTP requests using Burp Suite.
- Analyze HTTP requests, headers, and web traffic.
- Document security findings based on actual scan evidence.
- Practice responsible and authorized security testing.

3. NMAP: NETWORK SCANNING

Nmap (Network Mapper) is an open-source network scanning tool used to discover hosts, scan ports, and identify services running on a network.

Commands Used

- nmap -sS testphp.vulnweb.com — TCP SYN scan
- nmap -sV testphp.vulnweb.com — service/version detection
- nmap -A testphp.vulnweb.com — advanced scanning

Key Observations

- The target host was detected as up.
- Scanned ports appeared filtered/no-response.
- Service detection was limited because TCP responses were not received.

Possible Reasons

- Firewall filtering
- Network ACL/security rules
- Intermediate network filtering
- Packet dropping

Conclusion: Network filtering prevented complete port and service identification.

```
└─$ nmap -sS testphp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-05 16:59 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.00061s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored stat
es.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.40 seconds
```

Figure 1. Nmap TCP SYN scan output.

```
└─$ nmap -sV testphp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-05 17:00 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.00053s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored stat
es.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.70 seconds
```

Figure 2. Nmap service/version detection output.

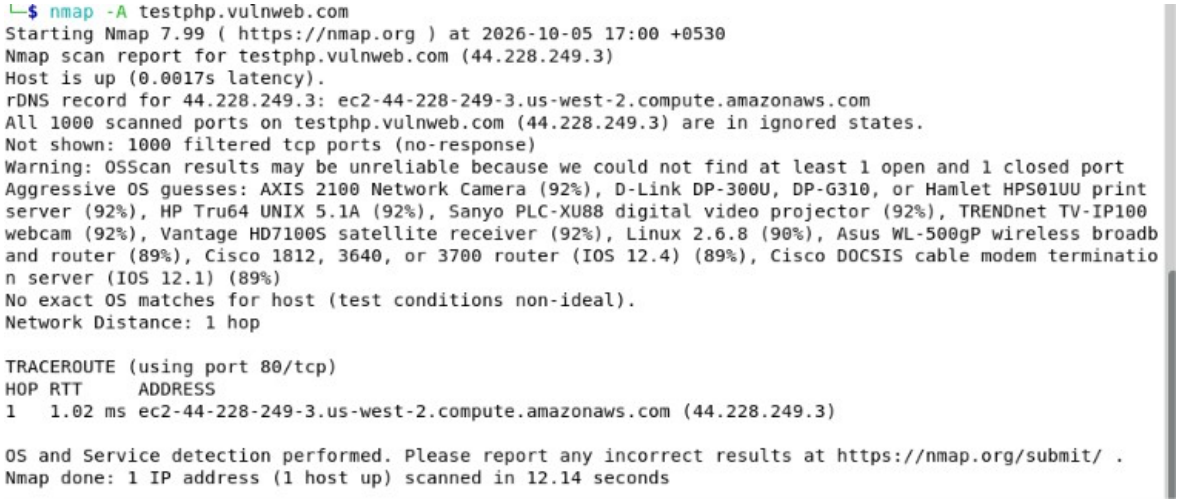


Figure 3. Nmap advanced scan (-A) output from the supplied presentation.

4. NIKTO: WEB VULNERABILITY SCANNING

Nikto is a web-server security scanner designed to detect potential security issues such as server misconfigurations, insecure files, outdated software components, and other commonly known web-server vulnerabilities.

Command Used: nikto -h http://testphp.vulnweb.com

- Nikto successfully resolved the target hostname.
- Multiple network addresses were identified for the specified target.
- However, 0 hosts were tested during the scan.
- Since no hosts were successfully tested, no web-server vulnerabilities were identified or reported by Nikto in this scan.
- The absence of reported vulnerabilities should not be interpreted as confirmation that the target is completely secure, because the scan did not successfully test a host.

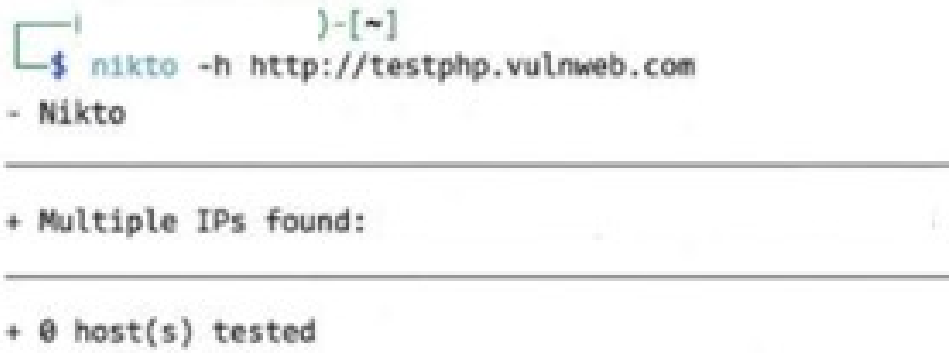


Figure 4. Nikto output showing target resolution and 0 hosts tested.

5. BURP SUITE: WEB APPLICATION SECURITY TESTING

Burp Suite is a web application security testing tool used to intercept and analyze HTTP/HTTPS traffic.

Proxy Configuration: 127.0.0.1:8080

- Configured the local proxy listener.
- Opened the authorized training website.
- Captured and inspected HTTP requests.
- Examined request methods, URLs, and headers.
- Reviewed HTTP History.
- Analyzed the captured request information.

Captured Request Information: HTTP Method, Request Path, Host, User-Agent, Accept Headers, and Connection Information.

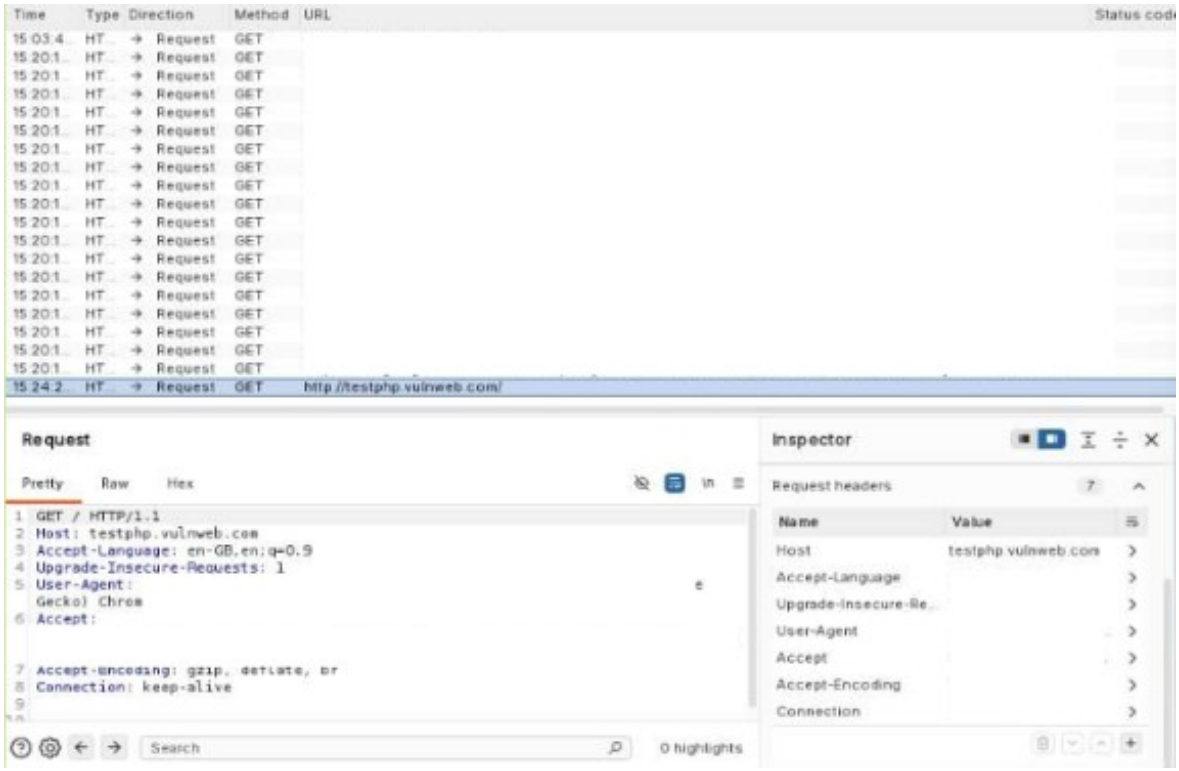


Figure 5. Burp Suite HTTP history and request inspection.

6. MINI VAPT TASK: FINDINGS & RESULTS

- Nmap — Filtered TCP Ports: The scanned TCP ports were identified as filtered, so Nmap could not determine whether they were open or closed.
- Nikto — Incomplete Web Scan: Nikto successfully resolved the target, but reported 0 hosts tested. Therefore, no web-server vulnerabilities could be confirmed.
- Burp Suite — HTTP Traffic Analysis: Burp Suite successfully captured and analyzed HTTP requests and headers, providing an understanding of web traffic and client-server communication.

Overall Interpretation: Network filtering limited the completeness of Nmap results; Nikto did not complete a host test; and Burp Suite provided usable HTTP request and header information for analysis.

7. CONCLUSION & LEARNING OUTCOMES

The Week 5 practical provided hands-on experience with three VAPT tools: Nmap, Nikto, and Burp Suite. The practical helped in understanding basic network and web application security testing.

- Understood the basic VAPT methodology.
- Performed network scanning using Nmap.
- Learned to interpret filtered TCP ports.
- Conducted a basic web-server assessment using Nikto.
- Understood the significance of incomplete scan results.
- Captured and analyzed HTTP requests using Burp Suite.
- Learned to examine HTTP headers and web traffic.
- Practiced basic security analysis and reporting.
- Understood the importance of testing only authorized targets.

8. REFERENCES

1. Nmap — https://nmap.org

2. Nikto — https://cirt.net/Nikto2

3. Burp Suite — https://portswigger.net/burp

4. OWASP — https://owasp.org

5. Acunetix / VulnWeb training target — http://testphp.vulnweb.com

Source note: The report content, commands, observations, conclusions, and screenshots are based on the supplied Week 5 PowerPoint. The supplied reference PDF was used as the formatting/model reference for the report structure.